

Perturbation Catalogue

An Integrated Resource for Exploring the Impact of Genetic Perturbations

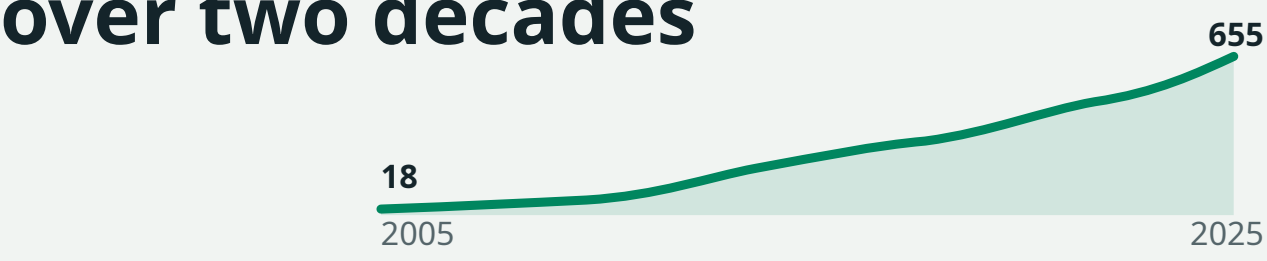
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Perturbation evidence is multiplying faster than it can be organised

A single pooled genome-scale Perturb-Seq dataset can **simultaneously profile tens of millions of cells across hundreds of thousands of perturbations**. That scale could reveal robust biology across studies, but identifiers, metadata, assays and processing pipelines differ. The Perturbation Catalogue aims to unify datasets across modalities and provide a shared access path for this evidence.

36× growth in perturbation papers over two decades



Europe PMC title/abstract query; 24 Aug 2026.
Scale example: [FirstWord Pharma story 7139212](#).

Our approach turns scattered datasets into a unified resource

1,237
datasets

19,235
targets

36
tissues

34
cell types

1,200
cell lines

169
diseases

CRISPR screens · 1,201 datasets

Perturbed gene → phenotype / fitness changes.

Perturb-Seq · 26 datasets

Perturbed gene → scRNA-seq, cell-wide transcriptome changes.

MAVE · 10 datasets

Combinatorial variant library → sequence-function maps.

We extensively curate metadata...

84
meta-
data
fields

Study & provenance

Title, authors, year, URI, licence

Model system

Organism, tissue, cell line, disease

Ensembl gene IDs

Perturbation & library

Targets, guides, library, variants

Readout & processing

Assay, technology, kit, reference

Ontology term mapping

And reprocess data from scratch

SRA / FASTQ

Raw reads + study-specific probes

→ **Cell × gene counts**

Shared reference + gene annotation

→ **Guide assignment**

Map cells to perturbations

→ **QC + filtering**

Final analysis-ready H5AD

Five Perturb-Seq datasets are live. Perturbation-control comparisons feed DEA and Hallmark GSEA. More datasets and modalities coming soon.

Browse the Catalogue

Search → Context → Filter → Explore

ebi.ac.uk/perturbation-catalogue

Perturbation Catalogue

Targets Datasets API Documentation About Request dataset

Perturbation Catalogue

Targets Search by metadata fields Search

Try searching for:

Perturbation Catalogue is a curated database that brings together data from various genetic perturbation experiments, including Perturb-Seq, CRISPR and MAVE screens, making it easier for researchers to study how modifying genes or proteins affects biological function across various biological and molecular contexts.

Perturbation Catalogue at a glance

Explore high-level statistics across the catalogue before diving into specific targets.

DATASETS 1237 (spanning 2014 - 2025)	TARGETS 19235	UNIQUE TISSUES 36	UNIQUE CELL TYPES 34
UNIQUE CELL LINES 1200	UNIQUE DISEASES 169		

Target: BRCA2

CRISPR screen data
MAVE data
Perturb-Seq (Perturbed)
Perturb-Seq (Affected)

Dataset: Nadig 2025 Jurkat

Study Title: Transcription-wide analysis of differential expression in perturbation screens
First Author: Alex Nadig
Last Author: Luke J. O'Connor
Year: 2020
Experiment Title: Jurkat day 7 essential gene set Perturb screen experiment
Data Modality: Perturb-Seq
Perturbation Type: CRISPR

Access records via API

Structured JSON for custom pipelines, reproducible analysis and machine learning.

GET /v1/perturb-seq/nadig_2025_jurkat/search?perturbation_gene_name=TFAM&limit=1

```
{  "perturbation": {    "perturbed_target_ensg": "ENSG00000108064",    "gene_symbol": "TFAM"  },  "effect": {    "effect_gene_ensg": "ENSG00000210082",    "gene_symbol": "MT-RNR2",    "log2Tc": -2.682  } }
```

Download entire datasets

Data is available in **Parquet** and **CSV.gz**, and metadata as **JSON**.

Stay tuned for more exciting things coming soon

Extract structured metadata faster

AI-assisted drafting with curator review and validated ontology mapping.

Reanalyse more designs

More kits, experimental structures and independent quality metrics.

Publish regular, versioned releases

Explicit provenance and stable snapshots for reproducible downstream use.

Reanalyse MAVE data

Enrich variant-to-function coverage with harmonised scores.

Visit: ebi.ac.uk/perturbation-catalogue Contact: perturbation-catalogue-help@ebi.ac.uk

